Partial Progress on MMS Guarantees of EFX Pure Nash Equilibria for Two Agents

Cleaned-up verification note

June 23, 2026

Abstract

We study the two-agent question, arising from the work of Amanatidis, Birmpas, Fusco, Lazos, Leonardi, and Reiffenhäuser, of determining the best maximin-share guarantee forced at pure Nash equilibria of no-money allocation mechanisms whose equilibria are always EFX. This note does not solve that problem. It gives partial progress: a menu-cone necessary condition for candidate equilibria, a sharp allocation-level restriction on non-MMS EFX outcomes, a cross-instance contradiction criterion that rules out a broad class of non-MMS equilibria and recovers the exact two-thirds obstruction, and a fragmented family explaining why this method does not by itself prove MMS.

Status. The results below are partial progress, not a solution of the open problem. In particular, they do not prove that every such EFX-PNE mechanism has only MMS equilibria, and they do not construct a mechanism with a non-MMS EFX equilibrium. The remaining difficulty is isolated in Section 5.

1 Model and target question

Let M be a finite set of indivisible goods and let the agents be 1 and 2. Each agent i has a nonnegative additive valuation $v_i : 2^M \to \mathbb{R}_{\geq 0}$, identified with its singleton values and extended additively. An allocation is an ordered partition $A = (A_1, A_2)$ of M.

For an agent i, the two-way maximin share is

$$\mu_i(v_i) = \max_{P \sqcup Q = M} \min\{v_i(P), v_i(Q)\}.$$

When the valuation is clear, we write simply μ_i. An allocation is α-MMS if

$$v_i(A_i) \geq \alpha\mu_i \qquad \text{for } i = 1, 2.$$

For two agents, A is EFX from agent i's perspective if, writing $j \neq i$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \qquad \text{for every } g \in A_j \text{ with } v_i(g) > 0.$$

The allocation is EFX if this holds from both agents' perspectives.

A deterministic direct-report no-money mechanism is a map

$$\mathcal{M} : (\mathbb{R}^M_{\geq 0})^2 \longrightarrow \{(S, M \setminus S) : S \subseteq M\}.$$

For a bid profile $b = (b_1, b_2)$, write $\mathcal{M}_i(b)$ for the bundle assigned to agent i. Given true valuations $v = (v_1, v_2)$, the bid profile b is a pure Nash equilibrium, abbreviated PNE, if for every agent i and every alternative bid b'_i,

$$v_i(\mathcal{M}_i(b_i, b_{-i})) \geq v_i(\mathcal{M}_i(b'_i, b_{-i})).$$

Definition 1.1 (Admissible EFX-PNE mechanism)**.** A mechanism $\mathcal{M}$ is called admissible if, for every two-agent additive instance v, it has at least one PNE and every PNE outcome is EFX with respect to the true valuations.

Problem 1.2 (EFX-PNE MMS guarantee)**.** Determine the best universal constant α^* such that every PNE outcome of every admissible mechanism is α^*-MMS. In particular, determine whether $\alpha^* = 1$.

This is the question left open in the discussion of Amanatidis et al. [1] and recorded in the open-problem entry [2]. The nonstrategic fact that every two-agent EFX allocation is 2/3-MMS is tight. The strategic question is whether the requirement that an allocation arise as a PNE of an admissible mechanism forces a stronger guarantee, possibly the full MMS guarantee.

2 Menus and robust EFX cones

Fix a mechanism $\mathcal{M}$, a bid profile $b = (b_1, b_2)$, and the outcome $A = \mathcal{M}(b)$. Against the opponent's bid, agent i has the unilateral menu

$$\mathcal{F}_i(b_{-i}) = \{\mathcal{M}_i(b'_i, b_{-i}) : b'_i \in \mathbb{R}^M_{\geq 0}\} \subseteq 2^M.$$

The cone of valuations for which A_i is a best response in this menu is

$$C_i(b, A) = \{w \in \mathbb{R}^M_{\geq 0} : w(A_i) \geq w(S) \text{ for every } S \in \mathcal{F}_i(b_{-i})\}.$$

Lemma 2.1 (Robust EFX cone)**.** *Let $\mathcal{M}$ be admissible. Suppose b is a PNE for $v = (v_1, v_2)$ and $A = \mathcal{M}(b)$. Then for every pair of valuations*

$$(w_1, w_2) \in C_1(b, A) \times C_2(b, A),$$

the same bid profile b is a PNE for the true valuation profile (w_1, w_2), and therefore A is EFX with respect to (w_1, w_2). Consequently, for each agent i and every $w \in C_i(b, A)$,

$$w(A_i) \geq w(A_j \setminus \{g\}) \qquad \textit{for every } g \in A_j \textit{ with } w(g) > 0,$$

where $j \neq i$.

Proof. If $w_i \in C_i(b, A)$, then $w_i(A_i)$ is at least the value of every bundle agent i can obtain by a unilateral deviation against b_{-i}. Thus neither agent can profitably deviate from b under the true valuations (w_1, w_2), so b is a PNE for that profile. Since $\mathcal{M}$ is admissible, every PNE outcome is EFX for the true valuations, and hence A is EFX with respect to (w_1, w_2). Taking $w_i = w$ and $w_j = v_j$, noting that $v_j \in C_j(b, A)$ because b is a PNE for v, gives the one-agent conclusion. □

Corollary 2.2 (Hidden maximin sides)**.** *Let $\mathcal{M}$ be admissible, let b be a PNE for v, and set $A = \mathcal{M}(b)$. If $v_i(A_i) < \mu_i$, then no side of any maximin partition for agent i belongs to the menu $\mathcal{F}_i(b_{-i})$.*

Proof. Let (P, Q) be a partition attaining μ_i. Then $v_i(P) \geq \mu_i$ and $v_i(Q) \geq \mu_i$, so both values are strictly larger than $v_i(A_i)$. If either P or Q belonged to $\mathcal{F}_i(b_{-i})$, agent i could deviate to obtain that bundle, contradicting that b_i is a best response. □

Corollary 2.2 is elementary but important: in any non-MMS equilibrium, every maximin-improving side must be absent from the deviating agent's menu. The next sections explain what this hiding requirement implies and where it fails to yield a full proof.

3 Allocation-level rigidity below MMS

The following lemma is independent of strategic behavior. It gives a sharp restriction on the number of positively valued goods in the envied bundle of a non-MMS EFX allocation.

Lemma 3.1 (Positive goods in the envied bundle)**.** *Let $A = (S, T)$ be EFX from the perspective of a single additive agent with valuation v. Normalize $\mu(v) = 1$, and put*

$$s = v(S) < 1, \qquad t = v(T), \qquad T^+ = \{g \in T : v(g) > 0\}, \qquad k = |T^+|.$$

Then $k \geq 2$ and

$$s \geq \frac{2(k-1)}{2k-1}.$$

Equivalently, if $s < 2r/(2r+1)$, then $|T^+| \leq r$. In particular,

$$s < \frac{4}{5} \Longrightarrow |T^+| = 2, \qquad s < \frac{6}{7} \Longrightarrow |T^+| \leq 3.$$

Proof. Since $\mu(v) = 1$, there exists a partition of M into two bundles each of value at least 1. Hence $v(M) = s + t \geq 2$, and so

$$t \geq 2 - s. \tag{1}$$

If $k = 0$, then $v(M) = s < 1$, impossible. If $k = 1$, then every partition of M has one part that excludes the unique positive good in T; that part has value at most $s < 1$, contradicting $\mu(v) = 1$. Thus $k \geq 2$.

Let x be a minimum-valued good in T^+. Since the positive goods of T have total value t, we have $v(x) \leq t/k$. EFX gives

$$s \geq v(T \setminus \{x\}) = t - v(x) \geq t - \frac{t}{k} = \frac{k-1}{k}t.$$

Using (1),

$$s \geq \frac{k-1}{k}(2-s).$$

Rearranging gives $s(2k-1) \geq 2(k-1)$, which is the desired bound. The equivalent form follows by replacing k with $r + 1$. □

Corollary 3.2 (Generic two-agent EFX bound)**.** *Every two-agent EFX allocation is 2/3-MMS.*

Proof. Fix an agent. If her allocated value is at least her MMS, there is nothing to prove. Otherwise normalize her MMS to 1 and apply Lemma 3.1. Since the envied bundle has at least two positively valued goods, the displayed lower bound gives $s \geq 2/3$. □

Remark 3.3 (Sharpness of Lemma 3.1). For every $k \geq 2$, the bound is tight at the allocation level. Let $d = 2k - 1$. Give the agent k goods of value $2/d$ and $2k - 2$ goods of value $1/d$. Allocate all $2k - 2$ small goods to S and the k larger goods to T. Then

$$v(S) = \frac{2(k-1)}{d}, \qquad v(T) = \frac{2k}{d},$$

and after removing any large good from T the remaining value is exactly $v(S)$. Thus the allocation is EFX from this agent's perspective. The total value is 2, and a partition into two bundles of value 1 is obtained by putting one large good and $2k - 3$ small goods on one side, and the remaining $k - 1$ large goods and one small good on the other side. Hence the MMS is 1.

4 A cross-instance contradiction criterion

The next result is the main partial-progress statement. It gives a sufficient condition under which a non-MMS EFX-PNE outcome cannot occur. The condition asks for two positively valued goods x, y in the other bundle and two positively valued witness goods p, q in the agent's own bundle such that each relevant mixed pair already exceeds the agent's equilibrium value.

Lemma 4.1 (The four-marker valuation)**.** *Let $0 < \eta < 1/3$. Consider four positive goods x, y, p, q with values*

$$w(x) = 1 + 2\eta, \qquad w(y) = 1, \qquad w(p) = 1, \qquad w(q) = \eta,$$

and let all other goods, if any, have value 0. In any two-agent EFX allocation for the identical valuation profile (w, w), each agent receives value at most 2. More precisely, ignoring zero-valued goods and swapping the two agents if necessary, the positive goods are split as

$$\{y, p\} \quad \textit{and} \quad \{x, q\}.$$

Proof. Zero-valued goods do not affect values or the positive-good EFX constraints, so it is enough to classify the partition of $\{x, y, p, q\}$. Look at the side containing x. There are eight possibilities.

If the x-side is $\{x, q\}$, the other side is $\{y, p\}$. The values are $1 + 3\eta$ and 2. Since $\eta < 1/3$, the first value is less than 2; the EFX inequalities are immediate because removing y or p from $\{y, p\}$ leaves value $1 \le 1 + 3\eta$, while removing x or q from $\{x, q\}$ leaves value at most $1 + 2\eta < 2$.

All other possibilities fail EFX. If the x-side is $\{x\}$, the other side is $\{y, p, q\}$; after removing q, the other side still has value $2 > 1 + 2\eta$. If the x-side is $\{x, y\}$, the other side is $\{p, q\}$; after removing y from the x-side, value $1 + 2\eta$ remains, which is larger than $1 + \eta$. The case $\{x, p\}$ is symmetric. If the x-side is $\{x, y, q\}$, the other side is $\{p\}$; after removing y, value $1 + 3\eta$ remains, larger than 1. The case $\{x, p, q\}$ is symmetric. If the x-side is $\{x, y, p\}$, the other side is $\{q\}$; after removing x, value 2 remains, larger than η. Finally, if the x-side is all four positive goods, the empty side plainly violates EFX. Thus only the split $\{x, q\} \mid \{y, p\}$ survives. $\square$

Proposition 4.2 (Two-marker contradiction criterion)**.** *Let $\mathcal{M}$ be admissible. Suppose b is a PNE for $v = (v_1, v_2)$, let $A = \mathcal{M}(b)$, and suppose that agent 1 is non-MMS at this outcome:*

$$s := v_1(A_1) < \mu_1.$$

Assume there are four distinct goods x, y, p, q such that

$$x, y \in A_2, \qquad p, q \in A_1,$$

all four have positive v_1-value, and

$$\begin{gathered} v_1(x) + v_1(y) > s, \\ v_1(x) + v_1(p) > s, \quad v_1(x) + v_1(q) > s, \\ v_1(y) + v_1(p) > s, \quad v_1(y) + v_1(q) > s. \end{gathered} \tag{1}$$

Then such a PNE cannot exist.

Proof. Choose $0 < \eta < 1/3$ and define the auxiliary valuation w by

$$w(x) = 1 + 2\eta, \qquad w(y) = 1, \qquad w(p) = 1, \qquad w(q) = \eta,$$

with $w(g) = 0$ for all other goods. Consider the identical auxiliary instance (w, w). Since $\mathcal{M}$ is admissible, this instance has at least one PNE; call it $b^* = (b_1^*, b_2^*)$. Its outcome is EFX, so Lemma 4.1 implies that agent 2's equilibrium value at b^* is at most 2.

Now form the mixed bid profile (b_1^*, b_2): agent 1 uses her auxiliary-equilibrium bid, while agent 2 uses her original-equilibrium bid. Let

$$(C, D) = \mathcal{M}(b_1^*, b_2),$$

where C is the bundle assigned to agent 1 and $D = M \setminus C$ is the bundle assigned to agent 2. Since b was a PNE for the original valuation profile, agent 1 cannot improve by deviating from b_1 to b_1^* against the fixed opponent bid b_2. Hence

$$v_1(C) \leq v_1(A_1) = s. \tag{2}$$

We now use the strict inequalities in (1). If $x \in C$, then none of y, p, q can lie in C, because x paired with any of them already has v_1-value larger than s, contradicting (2). Thus $\{y, p, q\} \subseteq D$, and

$$w(D) \geq w(y) + w(p) + w(q) = 2 + \eta > 2.$$

If $y \in C$, then none of x, p, q can lie in C, so $\{x, p, q\} \subseteq D$, and

$$w(D) \geq w(x) + w(p) + w(q) = 2 + 3\eta > 2.$$

Finally, if neither x nor y lies in C, then $\{x, y\} \subseteq D$, and

$$w(D) \geq w(x) + w(y) = 2 + 2\eta > 2.$$

The case $x, y \in C$ is already impossible by (1) and (2). Therefore, in all cases,

$$w(D) > 2. \tag{3}$$

Return to the auxiliary PNE b^*. If agent 2 deviates from b_2^* to the bid b_2 while agent 1 keeps b_1^*, she receives exactly the bundle D. By (3), this deviation gives her value greater than 2, whereas her value at the auxiliary PNE is at most 2. This contradicts the best-response condition at b^*. Hence the assumed original PNE cannot exist. □

Corollary 4.3 (No exact two-thirds non-MMS PNE)**.** *Let $\mathcal{M}$ be admissible. No PNE outcome A of $\mathcal{M}$ can satisfy*

$$v_i(A_i) = \frac{2}{3}\mu_i < \mu_i$$

for any agent i. Equivalently, no positive-MMS agent can attain the exact non-MMS ratio $2/3$ at such an equilibrium.

Proof. Suppose, for contradiction, that for agent 1 we have

$$v_1(A_1) = \frac{2}{3}\mu_1 < \mu_1.$$

Normalize $\mu_1 = 1$ and write $s = v_1(A_1) = 2/3$. Since the outcome is EFX, Lemma 3.1 applies to $S = A_1$ and $T = A_2$. It gives $|A_2^+| = 2$, because $s < 4/5$ and $|A_2^+| \geq 2$. Let these two positive goods be x and y.

The proof of Lemma 3.1 also gives $v_1(A_2) \geq 2 - s = 4/3$. On the other hand, EFX gives

$$v_1(x) \leq s = 2/3, \qquad v_1(y) \leq s = 2/3,$$

so $v_1(A_2) = v_1(x) + v_1(y) \leq 4/3$. Hence equality holds throughout and

$$v_1(x) = v_1(y) = 2/3. \tag{4}$$

Because $v_1(A_1) = 2/3 < \mu_1 = 1$, the bundle A_1 must contain at least two positive goods. Indeed, if A_1 contained at most one positive good, then no partition of M could place value at least 1 on both sides, contradicting $\mu_1 = 1$. Choose two positive goods $p, q \in A_1$.

By (4),

$$v_1(x) + v_1(y) > s,$$

and since p, q are positive,

$$v_1(x) + v_1(p) > s, \quad v_1(x) + v_1(q) > s, \quad v_1(y) + v_1(p) > s, \quad v_1(y) + v_1(q) > s.$$

Thus the hypotheses of Proposition 4.2 are satisfied, a contradiction. □

Remark 4.4. Together with the known theorem that every two-agent EFX allocation is 2/3-MMS, Corollary 4.3 recovers the qualitative conclusion that the exact tight nonstrategic ratio cannot be attained at a PNE of an admissible mechanism. The corollary does not supply an explicit better universal constant and does not imply MMS.

5 The fragmentation obstruction

The previous criterion explains why the proof does not automatically extend from 2/3 to 1. Just above 2/3, the extra value needed to form maximin bundles may be spread across many small goods, so no two individual own-bundle goods need satisfy the marker inequalities in Proposition 4.2.

Example 5.1 (Fragmented supplements)**.** Fix $s \in (2/3, 1)$ and set

$$a = 1 - \frac{s}{2}.$$

Let there be two goods x, y of value a and a collection R of small goods of total value s. Split R into two subcollections R_x and R_y of equal total value

$$v(R_x) = v(R_y) = \frac{s}{2} = 1 - a.$$

Consider the allocation

$$S = R, \qquad T = \{x, y\}.$$

Then $v(S) = s < 1$, while the partition

$$\{x\} \cup R_x, \qquad \{y\} \cup R_y$$

shows that $\mu = 1$. Moreover, the allocation is EFX from the perspective of this valuation because removing either x or y from T leaves value

$$a = 1 - \frac{s}{2} \leq s,$$

where the inequality is equivalent to $s \geq 2/3$.

If R is divided into sufficiently many tiny goods, then every individual $p \in R$ can be made to satisfy

$$v(p) \leq s - a = \frac{3s}{2} - 1.$$

In that case

$$v(x) + v(p) = a + v(p) \leq s,$$

so the marker inequalities of Proposition 4.2 fail. The MMS-improving bundles exist, but the supplementary value needed to pair with x and y is fragmented over many goods.

Example 5.1 is only an allocation-level obstruction. It is not a PNE construction, and it is not a counterexample to MMS for EFX-PNE mechanisms. Its role is to show that the two-marker proof strategy cannot by itself settle Problem 1.2.

6 A conditional MMS theorem under menu covering

The following condition is strong, but it clarifies exactly what kind of menu richness would force MMS.

Definition 6.1 (MMS-covering menu)**.** Fix a bid profile b with outcome A. The menu $\mathcal{F}_i(b_{-i})$ is MMS-covering for valuation v_i if, for every partition (P, Q) of M such that

$$v_i(P) \geq \mu_i \qquad \text{and} \qquad v_i(Q) \geq \mu_i,$$

at least one of P and Q belongs to $\mathcal{F}_i(b_{-i})$.

Theorem 6.2 (MMS under menu covering)**.** *Let $\mathcal{M}$ be any deterministic no-money mechanism, and let b be a PNE for valuation profile v with outcome A. If, for each agent i, the menu $\mathcal{F}_i(b_{-i})$ is MMS-covering for v_i, then A is MMS.*

Proof. Suppose A is not MMS. Then for some agent i,

$$v_i(A_i) < \mu_i.$$

Choose a maximin partition (P, Q) for v_i, so both sides have value at least μ_i. By the MMS-covering hypothesis, one of P, Q belongs to $\mathcal{F}_i(b_{-i})$. Agent i can therefore deviate to obtain a bundle of value at least μ_i, strictly improving on A_i. This contradicts the assumption that b is a PNE. □

7 What has and has not been proved

The verified progress is the following.

(i) Every candidate PNE outcome of a mechanism satisfying the EFX-equilibrium hypothesis satisfies the robust menu-cone condition of Lemma 2.1.

(ii) Every non-MMS EFX allocation with normalized value s has very few positive goods in the envied bundle when s is small; quantitatively, Lemma 3.1 gives $s \geq 2(k-1)/(2k-1)$.

(iii) Proposition 4.2 rules out non-MMS equilibria satisfying a concrete two-marker witness condition. This recovers the exact $2/3$ obstruction in Corollary 4.3.

(iv) Example 5.1 shows why the two-marker criterion does not settle the open problem: just above 2/3, the needed maximin supplement can be distributed among many tiny goods.

(v) Theorem 6.2 proves MMS under the additional assumption that equilibrium menus cover maximin partitions.

The missing implication is

$$\text{all PNE outcomes are EFX for every instance} \\ \Longrightarrow \\ \text{enough menu richness, or an equivalent substitute, to force MMS.}$$

This note does not establish that implication. Therefore it should be read as partial progress and a cleaned-up obstruction analysis, not as a solution of Problem 1.2.

References

[1] G. Amanatidis, G. Birmpas, F. Fusco, P. Lazos, S. Leonardi, and R. Reiffenhäuser. Allocating indivisible goods to strategic agents: pure Nash equilibria and fairness. *Mathematics of Operations Research*, 49(4):2425–2445, 2024. DOI: https://doi.org/10.1287/moor.2022.0058. Preprint: https://arxiv.org/abs/2109.08644.

[2] P. Nuti. Open Problems in Operations Research, entry W4389193896_p0. https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4389193896_p0&page=9&n=99.